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PI Name:	Brainard, George C. Ph.D.		
Project Title:	Optimizing Light Spectrum for Long Duration Space Flight		
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Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
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Space Biology Special Category:	None		
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Zip Code:	19107-5083	**Congressional District:**	1
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Glickman, Gena (Thomas Jefferson University) Smith, Peter (University of Arizona) Hanifin, John (Thomas Jefferson University) Pineda, Carissa (Thomas Jefferson University) Lockley, Steven (Brigham and Women's Hospital)		
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Task Description:	The goal of this project is to determine the optimal wavelengths of light for use as a countermeasure to sleep and circadian disruption for astronauts. Sleep deficits are considered a risk factor during space missions. The resulting physiological and behavioral changes from sleep and circadian disruption can threaten the success of a mission by diminishing alertness, cognitive ability, and psychomotor performance. Over 45% of all medications taken in space are sleep aids. Ambient light is the primary stimulus for circadian regulation in humans. While on Earth humans have a 24-hour day/night cycle to maintain healthy circadian entrainment. In space, astronauts must contend with either a rapidly changing or severely disrupted day/night cycle, and must work in a dimly lit spacecraft interior with few or no windows. Bright white light has already been implemented as a pre-launch countermeasure, but has yet to be used during space flight. Providing sufficient light intensities to work areas in the International Space Station (ISS), the planned Crew Exploration Vehicle (CEV), and future Lunar habitats raises several concerns, among which are heat production, energy consumption, and upmass. One potential way to mitigate these factors requires first a better understanding of how photic input regulates the human circadian system, and to characterize the effectiveness of this input for different wavelengths of light. From this data it may be possible to optimize the light spectrum both as a countermeasure for sleep and circadian disturbances before or during space exploration. Listed below are the specific aims of this four-year project. Given the recent changes in NASA priorities (Vision for Space Exploration) and subsequent budget cuts to this project, we have re-aligned the third specific aim: 1) Test the hypothesis that the white polychromatic fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for melatonin suppression when they are enriched in the blue portion of the spectrum. Our published action spectrum indicates that the blue monochromatic wavelengths (446-477 nm) are the most potent for melatonin suppression. This aim will determine if increasing the power in the blue region of a broad spectrum, polychromatic fluorescent light can increase potency for melatonin suppression while keeping the visual experience white. Working with Philips Lighting, we have expanded our polychromatic lighting exposure system. This year over 125 melatonin suppression trials have been completed with human subjects (over 175 trials since the project's conception in 2004.) 2) Test the hypothesis that the white fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for circadian phase-shifting when they are enriched in the blue portion of the spectrum. For this aim, we will test the magnitude of the phase-shift of circadian rhythms in healthy subjects following exposure to equal photon densities of blue-enriched fluorescent lights against the fluorescent lights currently used by NASA. This aim depends on the results from Aim 1, and will be initiated during the coming year. 3) Test the hypothesis that three ambient extraterrestrial light environments (skylight during low earth orbit for ISS and the CEV, ambient light on the Lunar surface, and ambient light on the surface of Mars) will differ in their capacity to regulate melatonin. Since circadian and sleep regulation depends on an astronaut's total exposure to both artificial and passive ambient illumination through windows or space suit visors, it is essential to determine the circadian effects of light exposures in non-terrestrial environments. To test this hypothesis, we are working collaboratively with the Lunar and Planetary Laboratory (LPL) investigators who have broad experience in characterizing the light spectra and intensities in a variety of extraterrestrial environments. Together, we are developing a polychromatic exposure system with the capacity to simulate the spectral characteristics of light in diverse extraterrestrial environments. We will use this apparatus to establish three melatonin suppression fluence-response curves with polychromatic light that simulates low Earth orbit, ambient light on the Lunar surface and ambient light on the surface of Mars. The first prototype of this system of optics and electronics has been successfully completed. In the coming year, we plan on completing the last fluence-response curve for melatonin suppression with blue enriched lamps. If the trials succeed, these lamps may provide NASA with their first deliverable lighting countermeasure to sleep and circadian disruption for use during space flight. We also plan to initiate subject recruitment to determine the circadian phase-shifting ability of prototype "blue enriched" fluorescent lamps. Finally, upon completion of the extraterrestrial light exposure system, we will initiate the melatonin suppression studies with a simulation of one of the three extraterrestrial lighting simulations. Ultimately, these data will impact NASA technology requirements to improve light treatment as a countermeasure for circadian and sleep-wake disruption in space flight, identify the best spectral transmission for space suit visors and space vehicle and habitat windows, and engineer the ideal spectral distribution for illumination of general living quarters. If the trials using blue enriched polychromatic lamps succeed, these lamps may provide NASA with their first deliverable lighting countermeasure to sleep and circadian disruption for use during space flight. Increasingly, however, it may be desirable to consider examining the potential utility of blue-enriched solid state luminaries for space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research conducted within the context of distributed interactive simulation models will provide the basis for developing effective patterns of communication and problem solving strategies as well as a range training procedures to enhance problem solving effectiveness. The Earth-based applications of this research will extend to the small operational group selection and training process as well as the management of stressful interactions and the maintenance of group cohesion and productivity. Not only can the outcome of these studies be expected to have an important impact on safety and the quality of life in many Earth-based applied settings, but larger societal units will ultimately benefit from the resulting conceptual and methodological advances that effectively promote social and ecological stability while concurrently enhancing an education and training technology that assures effective communication of an expanded generalizeable knowledge base. Aside from the Bioastronautics and Space Habitability programs that this project directly serves, the research audience for this NSBRI project is fairly broad. It applies to several areas of research in the basic sciences, including the circadian and neuroendocrine fields, where interest concerning the input physiology of the human circadian system is currently very strong. Psychologists and mental health professionals that use light therapy as an intervention for the treatment of affective and circadian related disorders such as SAD will be interested in this data as well. Strong interest also can be found in the light and lighting engineering fields, as knowledge of the impact of light on human health becomes better understood within the industry, and the notion of designing architectural illumination that addresses circadian health becomes more widespread.

Task Progress:

A significant portion of the global population suffer from chronic sleep loss and/or circadian-related disorders. Evidence for disease or illness to occur due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States alone, some 20 million Americans work shifts which do not allow them a biologically healthy nocturnal sleep cycle. This group has been shown to be more likely to suffer from a wide variety of ailments, including cardiovascular disease, gastrointestinal distress, cognitive and emotional problems. Furthermore, recent epidemiological studies of female night-shift nurses have shown that they are statistically more likely to suffer from breast cancer and colon cancer compared to day shift workers.
Aside from evidence for a breakdown in physical health, the effects of circadian disruption and sleep loss have long been known to have potentially dangerous behavioral effects. Mental fatigue, diminished alertness, loss of psychomotor coordination and physical performance are all commonly found in individuals with sleep loss, sleep debt, or circadian misalignment. The impact of these dangers affects many industries, including transportation, manufacturing, communications, and medicine. It has long been a source of concern for the military, as well. Additionally, many people experience the same effects due to air travel across several time zones.

Understanding the biological impact of ambient light on the human circadian system is a critical step toward the eventual application of countermeasures to these and other sleep and circadian-related disorders in civilian populations. Quantifying the relative potency of different wavelengths of light for melatonin suppression via the retinohypothalamic tract will provide key insights not only into the basic underlying physiology of this unique photosensory system, but also produce guidelines for how a misalignment of the system could be most effectively corrected. This research is also expected to have a wide-ranging impact on future types of artificial indoor lighting and architectural illumination in general. The lighting industry has taken notice of the biological impact different types of light is capable of having, and has already begun to invest in furthering our understanding of light's role for human health.

There are a number of existing therapeutic interventions using light that stand to benefit from an enhanced understanding of how specifically different wavelengths of the spectrum affect the human circadian system. A more efficient intervention, with increased potency and/or less side-effects could result. One such disorder currently being treated with bright white light is Seasonal Affective Disorder (SAD), also known as winter depression. It is estimated that as many as 1 in 5 Americans suffer from SAD or its milder version, subsyndromal Seasonal Affective Disorder (sSAD). Similar bright white light interventions are also used for treating jetlag. Side effects from exposure to bright white light for this and other therapies include: hypomania, headache, vision problems, nausea, dizziness, and anxiety. Optimizing the light spectrum for specific affective and/or circadian-related disorders could deliver the same medical impact with lower levels of light intensity, and potentially fewer side-effects.

Listed below are the specific aims of this four-year project. Given the recent changes in NASA priorities (Vision for Space Exploration) and subsequent budget cuts to this project, we have re-aligned the third specific aim:

1) Test the hypothesis that the white polychromatic fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for melatonin suppression when they are enriched in the blue portion of the spectrum. Our published action spectrum indicates that the blue monochromatic wavelengths (446-477 nm) are the most potent for melatonin suppression. This aim will determine if increasing the power in the blue region of a broad spectrum, polychromatic fluorescent light can increase potency for melatonin suppression while keeping the visual experience white. Working with Philips Lighting, we have expanded our polychromatic lighting exposure system. This year over 125 melatonin suppression trials have been completed with human subjects (over 175 trials since the project's conception in 2004.)

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3) Test the hypothesis that three ambient extraterrestrial light environments (skylight during low earth orbit for ISS and the CEV, ambient light on the Lunar surface, and ambient light on the surface of Mars) will differ in their capacity to regulate melatonin. Since circadian and sleep regulation depends on an astronaut's total exposure to both artificial and passive ambient illumination through windows or space suit visors, it is essential to determine the circadian effects of light exposures in non-terrestrial environments. To test this hypothesis, we are working collaboratively with the Lunar and Planetary Laboratory (LPL) investigators who have broad experience in characterizing the light spectra and intensities in a variety of extraterrestrial environments. Together, we are developing a polychromatic exposure system with the capacity to simulate the spectral characteristics of light in diverse extraterrestrial environments. We will use this apparatus to establish three melatonin suppression fluence-response curves with polychromatic light that simulates low Earth orbit, ambient light on the Lunar surface and ambient light on the surface of Mars. The first prototype of this system of optics and electronics has been successfully completed.

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